Boosting Sales at A.R Furniture through Strategic Inventory Optimization and Customer-Centric Design

A Proposal report for the BDM capstone Project

Submitted by:

Name: Abdul Hadi

Student Email: 22f1001445@ds.study.iitm.ac.in

Roll No: 22f1001445

IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

Declaration Statement

I am working on a Project titled "***Boosting sales at A.R Furniture through Strategic Inventory Optimization and customer centric design***". I extend my appreciation to, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analysed to assure its reliability.

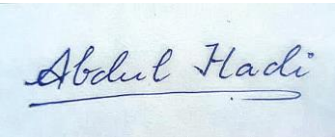
Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate:



Name: Abdul Hadi

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1. Executive Summary:

In today's competitive furniture market, A.R Furniture's is dealing with a few major problems like items going out of stock or staying unsold, customers not always finding what they're looking for, and rising competition from nearby stores. These issues are affecting sales and making it harder to keep customers happy.

The aim of this project is to help the business grow by using data to make better decisions. I plan to collect actual sales and stock data from the shop to see what items sell well and which ones don't. I will also conduct a simple survey to understand what customers really want and why they might not be fully satisfied. Some market research on competitors will also be included.

After collecting the data, I will clean and organize it using Excel, and then analyze it using Python for deeper insights if needed. I'll be using ABC classification to prioritize which products matter most, clustering to group customers based on their needs or behavior, and regression to understand what factors are affecting sales.

By doing this, I hope to give A.R Furniture's clear and practical suggestions like what to stock more of, how to target different types of customers, and how to compete more effectively. The final goal is to improve sales, reduce waste, and make the business more customer-friendly and competitive.

2. Organization Background:

A.R Furniture is a mid-sized furniture company that operates in both the B2B and B2C segments of the furniture industry. The company plays a dual role by supplying modular

furniture components and raw parts to local carpenters and small-scale furniture makers, while also offering a range of ready-made furniture products directly to end customers.

In the B2B segment, A.R Furniture acts as a vital resource for independent carpenters and interior contractors by providing them with essential materials like wood panels, fasteners, laminates, hinges, handles, and prefabricated modules. This not only supports local

craftsmanship but also positions A.R Furniture as a trusted supply partner in the regional market.

Simultaneously, the company serves the B2C market by designing and selling fully assembled, ready-to-use furniture pieces such as sofas, beds, tables, wardrobes, and modular kitchen units. With an emphasis on affordability and functional design, A.R Furniture targets middle-income households, offering a range of styles suited to urban living spaces.

Despite its strong presence in the local market, A.R Furniture is facing growing competition from large-format retailers, online furniture brands, and unorganized players. Additionally, challenges in inventory handling, fluctuating customer demands, and maintaining sales consistency have highlighted the need for strategic transformation. This project aims to address these operational and market challenges by leveraging data-driven insights to optimize inventory, improve customer engagement, and ultimately boost sales.

3. Problem Statement:

A.R Furniture, a company engaged in both supplying modular furniture parts to local carpenters and selling ready-made furniture to end customers, is facing critical business challenges in sustaining growth and competitiveness. The company operates in a highly fragmented and competitive furniture market, where both organized and unorganized players are exerting pressure on pricing, customer retention, and operational efficiency.

Despite having a diverse product portfolio and presence in both B2B and B2C segments,

A.R Furniture is struggling with the following interconnected problems:

3.1: Inventory Inefficiencies: Poor inventory visibility and inaccurate demand forecasting have led to frequent stockouts, overstocking, and rising holding costs, negatively impacting operational efficiency and customer fulfilment.

3.2: Limited Customer-Centric Approach: The current business strategy lacks effective integration of customer preferences and feedback, resulting in reduced customer satisfaction, low repeat purchases, and limited differentiation in the market.

3.3: Declining Sales Amid Competitive Pressure: Growing competition from large retailers and online furniture sellers is eroding A.R Furniture's market share. The absence of data-driven sales strategies and competitive benchmarking has further contributed to stagnating revenue growth.

These issues highlight the urgent need for a structured and strategic approach that combines inventory optimization, customer-centric product and service design, and data-informed sales planning to restore growth and secure A.R Furniture's position in the market.

4. Background of Problem:

A.R Furniture is a business that sells both ready-made furniture and furniture parts to local carpenters. In recent times, the company has faced several challenges due to increasing competition in the market. Many new furniture stores and online sellers are attracting customers by offering better services, faster delivery, and more choices.

One major problem for A.R Furniture is managing its inventory. Sometimes, the store runs out of items that are in high demand, while other times it has too much stock of items that don't sell quickly. This causes a waste of storage space and leads to loss of sales and profits.

Another issue is that customer needs and preferences are changing. A.R Furniture does not have a proper system to understand what customers want. As a result, they are unable to give a personalized shopping experience, which affects customer satisfaction and loyalty.

To stay ahead in the competitive market, A.R Furniture needs to improve how it manages its stock, understand its customers better, and use data to make smarter business decisions. This project will focus on solving these problems using data analysis and customer feedback to help the business grow and succeed in the long run.

5. Problem-Solving Approach:

To solve the problems A.R Furniture's is facing, I am going to follow a simple and practical step-by-step approach using real data. I will mostly use tools like Excel but if needed I use tool like Python and Power BI.

5.1 Data Collection

I will collect the following data:

- Sales records to understand which products sell well and which don't
- Stock and inventory logs to check for overstock or frequent stockouts
- Customer feedback through surveys to learn what buyers really want
- Competitor information like prices and popular products (via market research)

5.2 Tools Used

- Excel for data cleaning, charts, and ABC classification
- Python for deeper analysis like regression and customer grouping
- Power BI for creating dashboards to present insights clearly

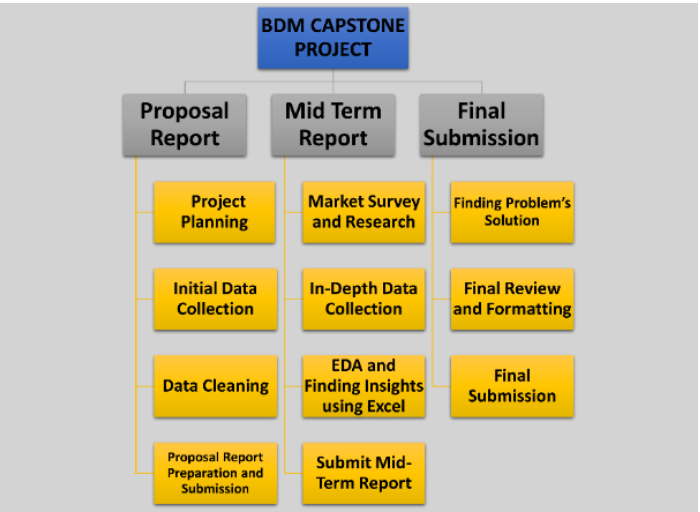
5.3 Solution Methods

Each problem will be solved using the right analysis:

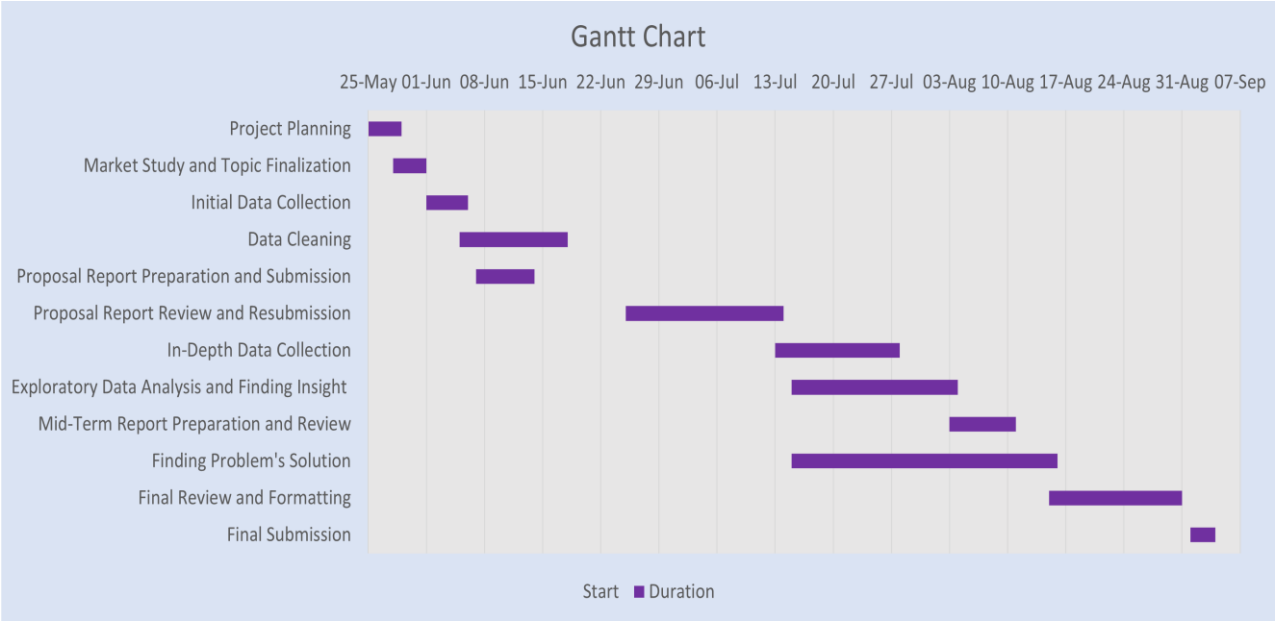
- Inventory issues: Use ABC classification to find fast- and slow-moving items
- Customer dissatisfaction: Use survey data + grouping (clustering) to tailor offerings
- Sales decline: Use regression to predict demand and plan better

6. Expected Timelines:

6.1 Work Breakdown Structure:



6.2 Gantt Chart:



7. Expected Outcome:

One of the key results will be a more optimized inventory system. This means fewer stockouts, less overstocking, and better planning of what products need to be available at what time. This will help reduce storage costs and increase profit margins.

Another important outcome will be insights into customer preferences. By analyzing data and customer feedback, I aim to help A.R Furnitures design and promote products that better match what customers actually want. This can improve customer satisfaction, loyalty, and repeat sales.

In addition, the project will provide suggestions for staying competitive in the market. This may include pricing strategies, marketing improvements, or product bundling ideas based on competitor analysis.

Overall, the project is expected to help A.R Furnitures boost its sales, reduce waste, and create a more customer-friendly business model, ultimately leading to long-term growth and stronger market presence.